Active Maintenance Report: galos_coppock_2023

Galos and Coppock (2023, Science Advances): Gender Composition Predicts Gender Bias

Table of contents

1 Summary **3**
- 1.1 Does the deposited archive run? . . . 3
- 1.2 Does the maintained rewrite reproduce the paper? . . . 4

2 Paper overview **4**

3 Original archive reproducibility **5**
- 3.1 The loop that cannot write its own output . . . 6
- 3.2 The bootstrap that ships at five iterations . . . 6

4 Errata **7**

5 Number-by-number comparison **9**

6 The extraction and the two instruments **11**

7 Tables 1 and 2 **13**

8 Figures **13**
- 8.1 Figure S1.1 . . . 15
- 8.2 Figures S4.5 to S4.61 . . . 16
- 8.3 Figure S5.62 . . . 16

9 Maintained rewrite **17**
- 9.1 Architecture . . . 18
- 9.2 Patterns replaced . . . 18

Archived at Zenodo: doi:10.5281/zenodo.24168385. This DOI always resolves to the latest release of the maintained version.

This repository holds the actively maintained replication code for Galos and Coppock (2023), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

Article	10.1126/sciadv.ade7979
Replication archive	10.7910/DVN/5MCRNY

The data are not redistributed here. The deposit is 19 files at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file against `original_manifest.csv`, which pins the file identifiers, folder paths, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not. The same script stops if `original/` holds anything the manifest does not list, since a stray file makes a working copy differ from a fresh clone in exactly the way a reproduction check exists to catch.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything, plus `in_text_claims.R`, which recomputes every claim the article makes and prints it beside the article's own sentence. `ground_truth/` ties every published number to the code that produces it and holds the extraction of the published pages. `original/` is created by the download script and is deliberately absent from the repository. `errata.qmd` renders `galos_coppock_2023_errata.pdf`, the corrections to the published article. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `galos_coppock_2023.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 19 files, produces every table and figure into `maintained/output/`, rebuilds the number-by-number comparison table in `ground_truth/` and its gates, and prints the recomputed value of every claim the article makes. Required

packages: tidyverse, metafor, broom, scales, rsample, modelsummary, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about two and a half minutes, almost all of it in the 500-resample bootstrap behind one standard error in the Results section.

A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The CSV, TeX and PNG files come back byte-identical. The eight figure PDFs always differ, because a PDF records the time it was written, so a changed hash there is not evidence of anything.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Almost. Eight of the ten deposited scripts run to completion on a current R installation with no changes at all. One fails outright and one runs but does not produce the published number.

`appendix_figures_and_loop.R` stops with an error on its first study. It binds a study's occupation-level estimates to a meta-regression row obtained from `broom::tidy()`, which carries no `fraction_female` column, and then sorts the combined frame with `fct_reorder(occupation_label, fraction_female)`. The meta row's ordering value is `NA`, and `forcats::lvls_reorder()` refuses to order a factor with a missing sort key. Because the error comes on the first iteration, the loop never reaches its last line, which is the one that writes `data/gender_gradients_df.rds`. That file is deposited and is correct, and re-deriving it from the occupation CATEs agrees with it to 4e-13 on every estimate and standard error, so nothing downstream is wrong. What is missing is any signal that the script and the object it is meant to write have come apart.

`analyze_cates.R` runs, but ships with `sims <- 5` where the paper used 500, and a comment saying so. Five resamples give a bootstrapped standard error of 0.54; five hundred give 0.578, which is the published 0.58. A reader who runs the archive as deposited gets a number that disagrees with the paper in the second decimal place, with no error and no warning.

Neither problem needs author intervention: the first is a two-line fix and the second is a one-character one. Every package the archive names still installs from CRAN.

Running the archive is not the same as reproducing the paper, and this archive is a clear case of the difference. The scripts cover Tables 1 and 2, Figs. 4 and 5, and five of the supplementary figures, though every one of them draws to the active graphics device and none calls `ggsave()`, so the deposit as shipped writes no figure files at all. They do not cover the 57 study-by-study panels of section S4, because the loop that draws them fails and its `ggsave()`

calls are commented out in any case, and they do not cover the study manifest of table S2.1, for which the deposit contains no code. That is 57 published figures and one published table with no deposited source.

1.2 Does the maintained rewrite reproduce the paper?

The ground truth records 533 values. 491 of the 519 that can be checked against the article reproduce, including all 57 study-level average treatment effects and their standard errors from fig. S1.1, all 12 survey experiment effects from fig. S3.4, all 37 study-level gender gradients from fig. S5.62, every coefficient, standard error, tau-squared and sample size in Tables 1 and 2, and all 335 estimate-and-standard-error labels printed on the 57 study panels.

The 28 that do not divide into two groups: 5 where the article disagrees with itself, and 23 where the deposit does not carry the quantity the published figure prints.

The article against itself. The paper reports that bias is larger in women-dominated than in men-dominated occupations "by a statistically significant 2.10 points". The archive and the rewrite both give 2.10547, which rounds to 2.11 at the printed precision; the published figure is the truncation. Published Table 2 prints the Model 2 intercept as −3.72 with no significance stars, where the coefficient's p-value is 0.00104 and the numerically identical −3.73 intercept in Model 3 does carry its two stars; the estimate and its standard error are right and only the marker is missing. Table S2.1 gives 18 occupations for Gonzalez et al. and 24 for Dahl and Krog, where the corresponding study panels, figs. S4.37 and S4.32, plot 17 and 21. The abstract says the experiments were carried out across five continents, where the paper's own evidence map is drawn as six continent facets and its earliest study is Australian.

The deposit against the figures. Every study panel labels each occupation with three lines: its name, the share of women in it, and a sample size. The first two reproduce for all 57 panels. The third is only partly recoverable, because the deposit carries no subject counts. What it does carry is the residual degrees of freedom of each occupation's own regression, and `df + 1` equals the published sample size exactly for 167 of the 298 occupations, which is every occupation of 35 of the 57 studies. For the other 22 the published figure is several times larger, which is what a design with several applications per job opening gives, and the deposit provides no way to recover the multiplier. The same gap shows up in table S2.1, whose subject counts reconstruct for the same 35 studies.

Not one of the 28 is a defect of the rewrite.

2 Paper overview

Citation: Galos, D. R. and Coppock, A. (2023). "Gender composition predicts gender bias: A meta-reanalysis of hiring discrimination audit experiments." *Science Advances*, 9, eade7979. DOI: 10.1126/sciadv.ade7979

Summary: Since 1983, more than 70 employment audit experiments have randomized the gender of fictitious job applicants, and the results point in both directions. This paper reconciles them with what the authors call a meta-reanalysis. The microdata for 57 studies reported in 51 papers are obtained or reconstructed, each study is reanalyzed to give conditional average treatment effects at the occupation level, occupational gender composition is merged in from International Labour Organization statistics, and the resulting 298 occupation-level estimates are meta-regressed on the share of women in the occupation. That slope is the gender gradient, and it is strongly positive: 8.15 percentage points per unit share female, with a standard error of 1.50. In a fully men-dominated occupation the expected effect of being a woman is −3.02 points; in a fully women-dominated one it is +5.13. Discrimination, in other words, preserves the status quo gender distribution rather than correcting it. The gradient survives country, year, and study fixed effects, holds separately among majority and minority status applicants, and reappears in a parallel meta-analysis of 12 vignette survey experiments.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0.

Script	Status on current R	Resolution
helpers.R	Clean (sourced by every analysis script)	No changes required
analyze_ates.R	Clean	No changes required
analyze_cates.R	Runs, but at 5 bootstrap resamples rather than the published 500	Set sims to 500, as the script's own comment instructs
analyze_cates_ethnicity.R	Clean	No changes required
analyze_cates_regression_tables.R	Runs; warns that redundant predictors were dropped from one model	No changes required; the warning is about collinear fixed effects
appendix_cates_survey_field.R	Clean	No changes required
appendix_evidence_map.R	Runs; warns that the size aesthetic for lines is deprecated	Use linewidth instead of size
appendix_figures_and_loop.R	Fails at fct_reorder on an NA ordering value	Replace the NA sort key before reordering
appendix_gender_gradient.R	Clean; reads the data file the failing loop script is meant to write	No changes required
appendix_survey_ates.R	Clean	No changes required

Every package the archive's README names still installs from CRAN: `tidyverse`, `metafor`, `broom`, `scales`, `texreg` and `xtable`. `texreg` is the one worth checking rather than assuming, because it is a table-formatting package that writes only to the console; version 1.39.5 was released in December 2025 and is actively maintained. `xtable` is loaded by `appendix_figures_and_loop.R` and never called.

The archive's own checks are reproduced by `ground_truth/archive_values.R`, which sources the deposited scripts unmodified, harvests the objects they assign and the LaTeX tables

they print, and writes `ground_truth/archive_values.csv`. That file is the source of the `value_script` column below. It is not part of `run_all.R`, because "does the deposit reproduce the paper" and "does the rewrite reproduce the paper" are different questions and answering the first one should not be a precondition for running the second.

3.1 The loop that cannot write its own output

`appendix_figures_and_loop.R` walks the 57 studies, fits a within-study meta-regression wherever a study distinguishes three or more occupations, and draws one appendix figure per study. Inside the loop it builds a plotting frame:

```
local_dat <- local_dat |> bind_rows(local_meta)
local_dat <- local_dat |>
  ungroup() |>
  mutate(occupation_label = fct_reorder(occupation_label, fraction_female))
```

`local_meta` comes from `tidy()` applied to an `rma.uni` fit, and a tidied model has no `fraction_female` column, so `bind_rows()` fills that cell with `NA`. `fct_reorder()` then asks `lvls_reorder()` to order a factor by a vector containing a missing value, which errors. Replacing the missing key with `-Inf` before reordering fixes it and puts the meta row first, which is where the published appendix figures show it.

The loop's `ggsave()` calls are commented out in the deposit, so the only thing it produces that anything else consumes is the last line, `write_rds(gender_gradients_df, "data/gender_gradients_df.rds")`. That file is deposited, and `appendix_gender_gradient.R` reads it to build fig. S5.62. The archive is therefore internally consistent while being unable to rebuild itself: the deposited object is right, and the code that made it does not run.

That last line is also a hazard for anyone checking the archive, because it writes into the archive directory and would replace a deposited file with the output of the run. Deposited code is run in a scratch copy under `tempdir()` here for that reason, and `original/` is verified against the manifest on every run.

3.2 The bootstrap that ships at five iterations

`analyze_cates.R` estimates the difference in the magnitude of bias between fully women-dominated and fully men-dominated occupations, and bootstraps its standard error:

```
set.seed(343)
# sims <- 500
sims <- 5
```

The published standard error is 0.58. At five resamples the archive prints 0.54.

4 Errata

Twelve things in the published article do not survive the comparison: seven statements and figures checked against the deposit, and five reference-list entries checked against Crossref or against the list's second printing in the supplementary materials. They are collected in `galos_coppock_2023_errata.pdf` at the root of this repository, which renders from `errata.qmd` and computes every corrected number from `maintained/output/` at the time it is rendered. The note also carries seven passages this edition reprints unchanged, each quoted with the reason it is not corrected. **None of them changes a conclusion of the paper.**

Table 2: The twelve corrections, in the order the errata note gives them. Both columns are read from the errata spine.

Correction	Evidence
The record spans six continents, not five	The paper's own evidence map, fig. S2.2, is drawn as six continent facets: Africa, Asia, Europe, N. America, Oceania, S. America. Section S2 names Oceania among the regions where the boom in audit experimentation occurred, and the earliest study in the whole record, the Australian one the introduction discusses, is on that continent. The sentence describes the wider literature rather than the 57 studies reanalysed here, but a superset of those studies cannot span fewer continents, so five is too few under any reading.
Table S2.1 overstates the occupation counts for two studies	The article disagrees with itself in these two rows. Figs. S4.37 and S4.32 plot 17 and 21 occupations for the same two studies, and 17 and 21 is what the deposited data hold. The other 55 rows of the column are right. The table is not reprinted here because its N subjects column cannot be regenerated from the deposit, which carries no subject counts.
The difference in the magnitude of the two predicted biases	The deposited code and the maintained rewrite both give 2.10547, which rounds to 2.11 at the two decimal places the sentence prints. The published figure is the truncation rather than a different number, and the bootstrapped standard error, 0.5781, is 0.58 as published.
Figure 4 colors imputed gender composition categories as though the study authors had given them	Of the 298 study-occupation cells the figure draws, only 174 carry a gender composition category the original study's own authors assigned; the remaining 124 were filled in by us from where the occupation falls in the ILO gender distribution. The caption discloses that in words, so the article is accurate about what it did, but the figure gives all 298 points the same three colors, and a color is a claim about the occupation that a reader cannot tell apart from the 174 that were coded. The corrected figure draws the 124 imputed cells in grey, keeping their position, their interval and their weight. NO ESTIMATE MOVES: the meta-regression is fitted on the continuous share of women in the occupation and never on the category, so Tables 1 and 2, the gender gradient and every study-level effect in the supplementary materials are exactly as published. What changes is which of 298 points is which color. The coding is the same authors' own, carried back from the successor project, where the distinction is kept as a column and the imputed categories are dropped rather than plotted.

The article cites a supplementary figure that does not exist	The supplementary materials list Figs. S1.1 to S6.63 in their front matter, and the PRISMA flow diagram is the last of them. There is no fig. S6.66. The same paragraph cites the evidence map as fig. S.2.2, where the figure itself is numbered S2.2.
Table 2's Model 2 intercept carries no significance marker	The Model 2 intercept is printed as −3.72 with no marker, at a p-value of 0.00104. Five decimals rather than the three the table's own note uses: at three the p-value reads 0.001, which is the threshold for the third star rather than the second, and the correction would be ambiguous inside the rule it is being compared against. The numerically identical Model 3 intercept, −3.73, does carry its two stars. The estimate and its standard error are right and only the marker is missing, so what is corrected here is the table rather than any sentence.
Figure S1.1 prints a clipped estimate for Neumark et al.	The figure's own text carries the right number and the page does not print it. Read off the glyphs, the row holds exactly 2 . 3 space (8 . 4) beginning at x0 = 268.50, which is where every other three-character value in the column begins, and no minus and no 1 is inked anywhere to its left: the short dash that sits there is the row's axis tick, which the positive rows carry too. The authors' own field_ates_plot.pdf is the same in both respects, so the leading −1 was lost when the figure was drawn rather than when it was composed: the label overflows the plot panel and its −1 is clipped out of the ink, while the text layer keeps it, which is why pdftotext reports −12.3 for a page that shows 2.3 and why a token gate cannot see this defect at all. Nothing about the estimate is in doubt. The article's own introduction prints it as (−12.3 points, SE: 8.4 points) and the deposit gives −12.307692 (8.384118), so the paper disagrees with itself the way entry 2 records for table S2.1. The corrected figure is the maintained pipeline's own, which draws the full value: of the 420 words and numbers it carries, 419 are on the published page and the one exception is the restored −12.3.
Reference 22 demotes half of an author's surname to an initial	The third author's surname is the compound Baier Manwell, and the entry reads its first element as a middle initial. Crossref, doi 10.1097/acm.0000000000000552: Carnes, Molly; Devine, Patricia G.; Baier Manwell, Linda; Byars-Winston, Angela. PubMed indexes her the same way, as Baier Manwell L.
Reference 65 breaks two surnames beginning "Mc"	Joanne E. McKenzie's surname is set with a space, and Steve McDonald's is split so that its "Mc" reads as a middle initial. Crossref, doi 10.1136/bmj.n71: McKenzie, Joanne E is the second author and McDonald, Steve the nineteenth. The defect is specific to these two names rather than general to the typesetting: the same entry sets L. A. McGuinness correctly three names later, and reference 60 sets M. McPherson correctly.
Reference 9 separates two coauthors with a semicolon	The entry names three coauthors and separates the first two with a semicolon and the second two with a comma, so it is inconsistent with itself before it is compared with anything. Crossref, doi 10.2307/2946676: David Neumark, Roy J. Bank and Kyle D. Van Nort are three authors of one article, Q. J. Econ. 111, 915-941 (1996). Every one of the other 127 entries in this list separates its authors with commas, and the supplementary materials set this same reference with a comma. The defect is specific to the article's printing of it.

Reference 71 in the supplementary materials drops the period after its number	Every other entry in both printings of this list numbers itself with a digit and a period, and the article's own printing of this same reference has one. Read off the line boxes of supplementary p. 53, reference 71 begins at the item edge x0 = 0.088 like its neighbours and its number is followed by a space rather than a period. It is the only one of the 128 that is, which is what makes it a slip rather than a convention.
Reference 83's page range ends one page early, in both printings of the list	Crossref, doi 10.1086/511799: Shelley J. Correll, Stephen Benard and In Paik, "Getting a Job: Is There a Motherhood Penalty?", American Journal of Sociology 112, 1297-1339 (2007). The supplementary materials reprint the entry with the same range, so it is corrected in both places.

Three of the twelve are contradictions between two parts of the article: the abstract against the paper's own evidence map, table S2.1 against the study panels, and the PRISMA cross-reference against the supplement's own figure numbering. Two more are a dropped typesetting marker and a truncation, both of which the deposited code itself gets right. Two are figures: Figure 4 colors imputed categories as though they had been coded, and Figure S1.1 prints an estimate clipped short of its own value. The rest are reference-list entries, which the deposit cannot speak to at all: one author's compound surname reduced to an initial, two surnames beginning "Mc" broken in a single entry, two defects found by diffing the article's printing of the list against the supplement's, and a page range that ends a page early in both.

This is separate from the question of whether the maintained rewrite corrected anything in the deposited code. It did not: the rewrite makes no analytical change and fixes no bug, and the two changes it does make to how the archive runs, raising the bootstrap from five resamples to the published five hundred and replacing an `NA` sort key, are described above.

5 Number-by-number comparison

The ground truth is at `ground_truth/galos_coppock_2023_ground_truth.csv` and is generated by `ground_truth/build_ground_truth.R`, which runs as the last step of `run_all.R`. Its only typed numbers are the `value_paper` column, read from the published article and its supplementary materials; `value_script` is read from the harvest of the deposited code and `value_rewrite` out of `maintained/output/`. The verdicts are computed at the precision the article prints rather than asserted.

It records 533 values. 299 of them have both a published counterpart and a deposited-code counterpart, and of those 293 match and 6 do not.

Most of the published values are inside figures rather than in prose. Figures S1.1, S3.4 and S5.62 print the estimate and standard error beside every point, which makes 218 published numbers that no sentence in the paper states. The 57 study panels of section S4 print another 335. All of them are extracted from the supplementary PDF and checked one by one.

Table 3: Where the checkable published numbers are, and how the rewrite does against them.

Location	Published values checked	Rewrite matches	Rewrite mismatches
Figures S4.5 to S4.61	173	151	22
Figure S1.1	116	115	1
Figure S5.62	76	76	0
Text	40	35	2
Figure S3.4	26	26	0
Table 2, Model 3	14	14	0
Section S5	8	8	0
Table 1, Model 1	8	8	0
Table 1, Model 2	8	8	0
Table 1, Model 3	8	8	0
Table 1, Model 4	8	8	0
Table 1, Model 5	8	8	0
Table 2, Model 1	8	8	0
Table 2, Model 2	8	7	1
Table S2.1	4	2	2
Section S1	3	3	0
Section S2	3	3	0
Section S4	2	2	0
Section S3	1	1	0

Table 4: Every published value the rewrite does not match, and where the fault lies.

Location	Quantity	Paper	Rewrite	Where the fault lies
Abstract	Continents the experiments were carried out across	5	6.000	paper_internal
Results, p. 6	Difference in magnitude, 100 vs 0 percent women (pp)	2.10	2.105	paper_internal
Table 2, Model 2	Intercept significance stars (0 = none, 1 = one, 2 = two, 3 = three)	0	2.000	paper_internal
Table S2.1	Subject counts reproduced	57	35.000	archive
Table S2.1	Occupation counts reproduced	57	55.000	paper_internal
Figure S1.1	ATE: Neumark et al. (9): United States	2.3	-12.308	paper_internal
Figure S4.8	Occupation sample size labels, Alden et al. (68): Sweden	2	0.000	archive
Figure S4.11	Occupation sample size labels, Asali et al. (70): Georgia	9	0.000	archive
Figure S4.13	Occupation sample size labels, Baert et al. (71): Belgium	1	0.000	archive
Figure S4.14	Occupation sample size labels, Baert et al. (72): Belgium	1	0.000	archive
Figure S4.17	Occupation sample size labels, Berson (75): France	1	0.000	archive
Figure S4.28	Occupation sample size labels, Carlsson (13): Sweden	13	0.000	archive
Figure S4.32	Occupation sample size labels, Dahl and Krog (84): Denmark	21	3.000	archive
Figure S4.33	Occupation sample size labels, Edo et al. (85): France	3	0.000	archive
Figure S4.37	Occupation sample size labels, Gonzalez et al. (89): Spain	17	0.000	archive
Figure S4.38	Occupation sample size labels, Hipp (90): Germany	1	0.000	archive
Figure S4.42	Occupation sample size labels, Neumark et al. (9): United States	1	0.000	archive
Figure S4.43	Occupation sample size labels, Nunley et al. (93): United States	6	0.000	archive
Figure S4.44	Occupation sample size labels, Oreopoulos (94): Canada	21	1.000	archive
Figure S4.46	Occupation sample size labels, Pedulla (96): United States, field experiment	4	0.000	archive
Figure S4.47	Occupation sample size labels, Petit (97): France	1	0.000	archive
Figure S4.50	Occupation sample size labels, Riach and Rich (8): Australia	7	0.000	archive
Figure S4.51	Occupation sample size labels, Riach and Rich (100): England	4	0.000	archive
Figure S4.52	Occupation sample size labels, Rivera and Tilcsik (101): United States, field experiment	1	0.000	archive
Figure S4.53	Occupation sample size labels, Rooth (102): Sweden	7	0.000	archive
Figure S4.54	Occupation sample size labels, Ruffle and Shtudiner (103): Israel	10	1.000	archive
Figure S4.60	Occupation sample size labels, Zhou et al. (108): China	4	0.000	archive

Table 4: Every published value the rewrite does not match, and where the fault lies. *(continued)*

Location	Quantity	Paper	Rewrite	Where the fault lies
Figure S4.61	Occupation sample size labels, Capéau et al. (82): Belgium	1	0.000	archive

The 11 remaining recorded quantities are ones the scripts produce and the paper never prints: the mean and median number of occupations per study, the count of survey occupation CATEs, the intercept and slope of the two fitted lines in fig. S3.3, which are drawn but never stated numerically, and the four distances that stand behind a sentence describing two quantities as similar or stable. They are recorded with no published counterpart rather than counted as matches.

A further 3 claims are recorded with a published value and no verdict, because the article states them as approximations. "Two-thirds (37 of 57)" and "over the past four decades" are true at the precision they claim and false at any finer one, so both numbers are recorded and neither is scored.

Coverage is recorded as a fraction, not as a yes or no. A float with one row against a hundred published cells satisfies any check that asks only whether the float appears, so the build counts the numbers each published float prints on its face and requires the ground truth to account for all of them. `ground_truth/float_coverage.csv` holds that count float by float. Of the 71 floats the article and its supplementary front matter list, 64 carry rows here. The 63 that carry rows and print numbers on their faces carry 1,389 published values between them, and the ground truth reaches every one, so each of those floats is covered in full. The build stops if any float falls below that without a stated reason.

The remaining 7 published floats have no row here, and the build stops if a float has neither a row nor a stated reason, since a false reason tells the next reader not to look. Figures 1 and 2 are diagrams of the two gender gradients the theory admits and plot no estimates. Figure 3 summarises the discrimination mechanisms the 51 papers discuss, coded by hand, and prints the six category counts beside its bubbles; the deposit carries no mechanism or discipline variable, so nothing in the pipeline can reach it. Figures 4 and 5 print no numbers on their faces, and the estimates and lines they draw are the 298 occupation CATEs and the Table 1 and Table 2 fits, all covered above. Figure S2.2 is a map of country-by-year tiles, and what it asserts is checked through the section S2 sentences that describe it. Figure S6.63 is a PRISMA flow diagram whose counts come from the screening process and not from any data file in the deposit.

6 The extraction and the two instruments

The comparison above answers "does the pipeline reproduce the published floats." It cannot answer "does the pipeline reproduce the published article," because a ground truth built float by float has no way to notice a sentence. Two further files close that gap.

The extraction. `ground_truth/published_claims.csv` is a hand-reviewed list of every numeric claim the article and its supplementary materials make: 596 of them, each with the string the page prints and a classification.

Table 5: The extraction, by claim type. A pipeline or descriptive claim must have both a block and a ground truth row; the other three types get a block only where the pipeline can reach the quantity, and are otherwise verified at the point of use.

Claim type	No block	Needs a block
definitional	6	12
descriptive	0	18
pipeline	0	500
structural	23	27
transcribed	8	2

The float cells are read off the supplementary PDF's own text layer by `ground_truth/extract_published_claims` which is not part of `run_all.R` because this repository does not carry the published PDFs. The prose claims, the two published tables and the front matter are typed from the page. That makes the extraction a second transcription of the same pages the ground truth transcribes, and the build compares the two: all 522 published values they share agree, digit for digit.

The second instrument. `maintained/in_text_claims.R` recomputes every claim the extraction says needs a block, from `maintained/output/`, by its own path. It carries the article's own sentence verbatim in a comment above each block and prints one line per claim. It never reads the ground truth. `build_ground_truth.R` runs it, counts what it printed against what the extraction declares, and stops if the two disagree; it then compares the two instruments value by value, and stops if they land on different numbers. The 175 claims about summarised floats are exempt from that last comparison, because their ground-truth value is a count of agreements with the published page and an instrument that never reads the page cannot recompute one.

Three findings came from the extraction and not from the floats. The abstract's "five continents" against an evidence map drawn as six is the first, and it is now the first entry of the errata. The second is the article's own cross-reference to fig. S6.66. The third is a disagreement the article has with itself about a date: the introduction says the earliest study sent its applications in 1986, while the abstract dates the record from 1983 and the deposit's own study table and evidence map both put that study in 1983. Which is right is a fact about the cited 1987 paper rather than about this deposit, so it is not corrected: it is entry 19, a note that quotes the sentence, prints no corrected side, and gives that reason.

A fourth is worth stating and is not an erratum either. The Design section says the PRISMA search "yielded over 70 papers that met our inclusion criteria", where the PRISMA flow diagram of fig. S6.63 reports 51 reports included and 89 assessed for eligibility. No reading of the diagram gives over 70, and nothing in the deposit records the search, so the sentence cannot

be corrected from the pipeline. It is entry 18, a note for that reason. Entry 5 quotes the same sentence to correct the figure it cites and reprints "over 70 papers" unchanged on both sides.

7 Tables 1 and 2

Both published tables reproduce in full, coefficients, standard errors, significance markers, tau-squared and sample sizes alike, with the single exception of the Model 2 intercept marker in Table 2.

Table 6: Table 1 as reproduced by the maintained rewrite. Standard errors in parentheses.

Model	Intercept	Gender gradient	Tau squared	Nobs
Model 1	-3.02 (0.81)	8.15 (1.50)	12.57	298
Model 2	-2.48 (1.58)	8.33 (1.58)	11.71	298
Model 3	-5.43 (1.91)	8.69 (1.56)	12.56	298
Model 4	-5.20 (1.82)	7.97 (1.57)	10.22	298
Model 5	1.05 (1.96)	7.51 (1.54)	8.89	298

Table 7: Table 2 as reproduced by the maintained rewrite. Model 1 is the majority status group, model 2 the minority status group, model 3 both. Standard errors in parentheses.

term	Model 1	Model 2	Model 3
Intercept	-1.55 (1.31)	-3.72 (1.13)	-3.73 (1.14)
Gender gradient	5.70 (2.50)	9.08 (2.26)	9.11 (2.28)
Majority status	NA	NA	2.19 (1.74)
Majority x gradient	NA	NA	-3.42 (3.37)

8 Figures

The main text carries two figures of results and the supplementary materials sixty-one more, of which the rewrite reproduces all sixty-one: the two forest plots, the evidence map, the survey-against-field comparison, the gender gradient forest plot, and the 57 study-by-study panels.

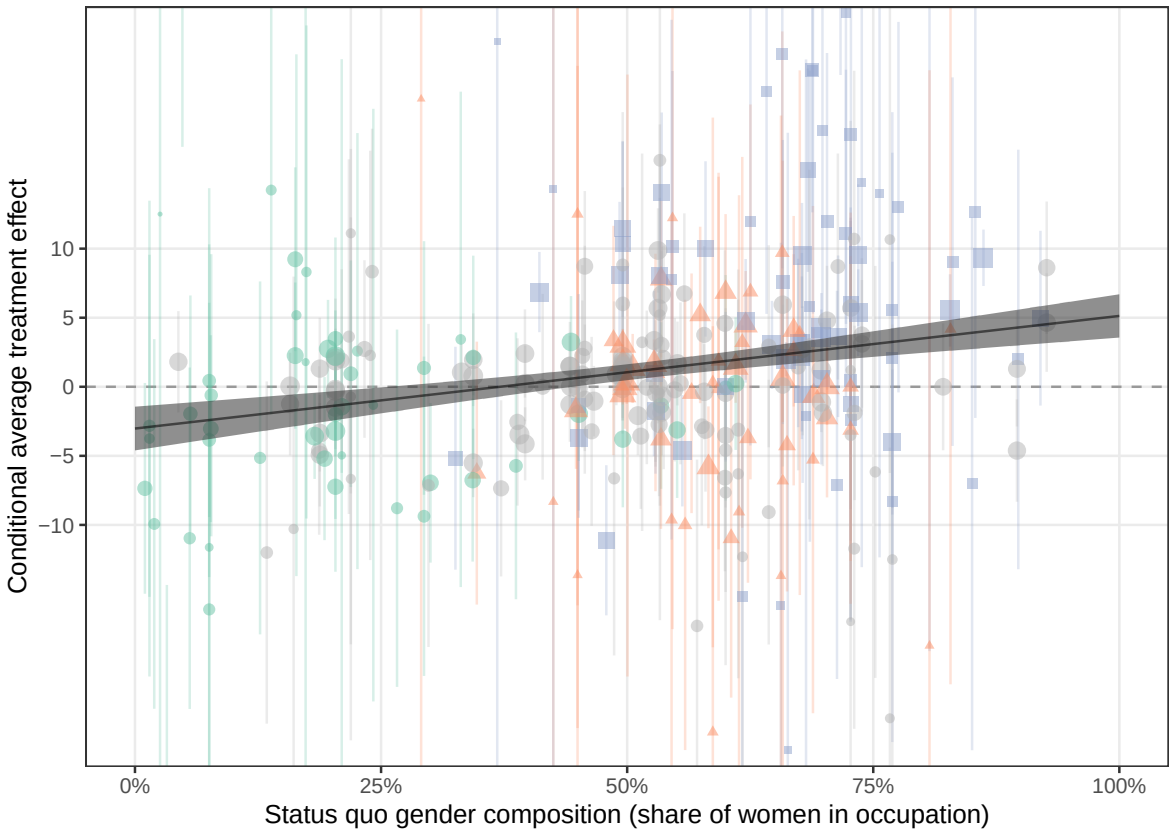


Figure 1: Figure 4: occupation-level CATEs against gender composition, with the Table 1 model 1 regression line.

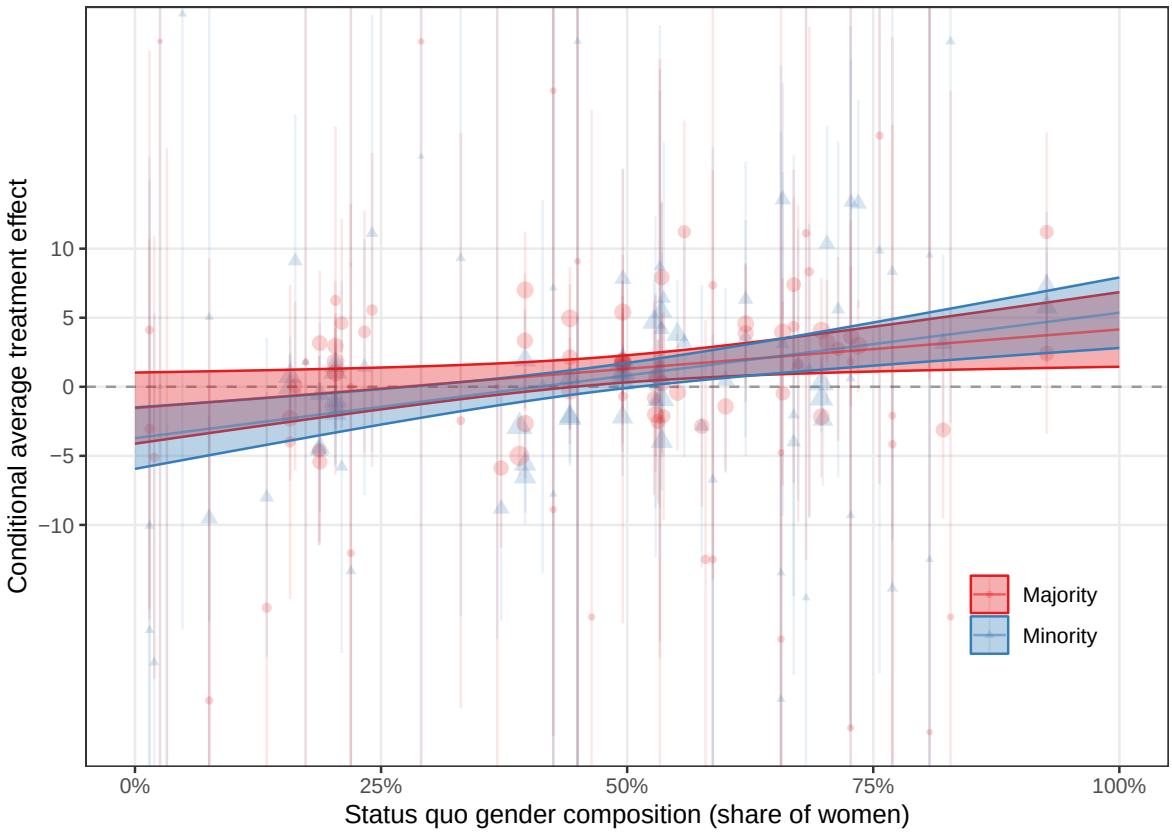


Figure 2: Figure 5: the same estimates split by majority and minority applicant status.

8.1 Figure S1.1

Fig. S1.1 is a forest plot of all 57 field studies, each labelled with its estimate and standard error. All 116 of those labels reproduce, along with the meta-analytic summary of 1.2 points with a standard error of 0.4.

Table 8: The three largest and three smallest of the 57 study-level effects in fig. S1.1, all of which reproduce.

Study	ATE (SE)
Albert et al. (67): Spain	9.9 (1.8)
Birkelund et al. (76): the Netherlands	8.3 (3.2)
Becker et al. (74): Austria, Germany, Switzerland	8.1 (0.9)
Rivera and Tilcsik (101): United States, field experiment	-3.8 (3.3)
Arai et al. (69): Sweden (equivalent CVs)	-6.3 (4.0)
Neumark et al. (9): United States	-12.3 (8.4)

8.2 Figures S4.5 to S4.61

The 57 study panels are the densest published numbers in the paper and the part of it the deposit does not draw. `figure_s4_study_panels.R` rebuilds them as one paginated PDF, one page per study, and writes every plotted value to CSV. All 335 estimate-and-standard-error labels and all 298 gender composition labels reproduce. The sample size labels are the exception described above: `df + 1` recovers 167 of the 298. That is every occupation of 35 studies, plus five scattered across the other 22.

8.3 Figure S5.62

The 37 study-level gender gradients are the paper's own evidence that a single audit experiment cannot see the pattern it documents: only 7 of the 37 are individually significant, which the paper glosses as putting the power of the average study at roughly 19 percent. All 37 estimates, their standard errors, and the meta-analytic 9.0 with a standard error of 2.1 reproduce.

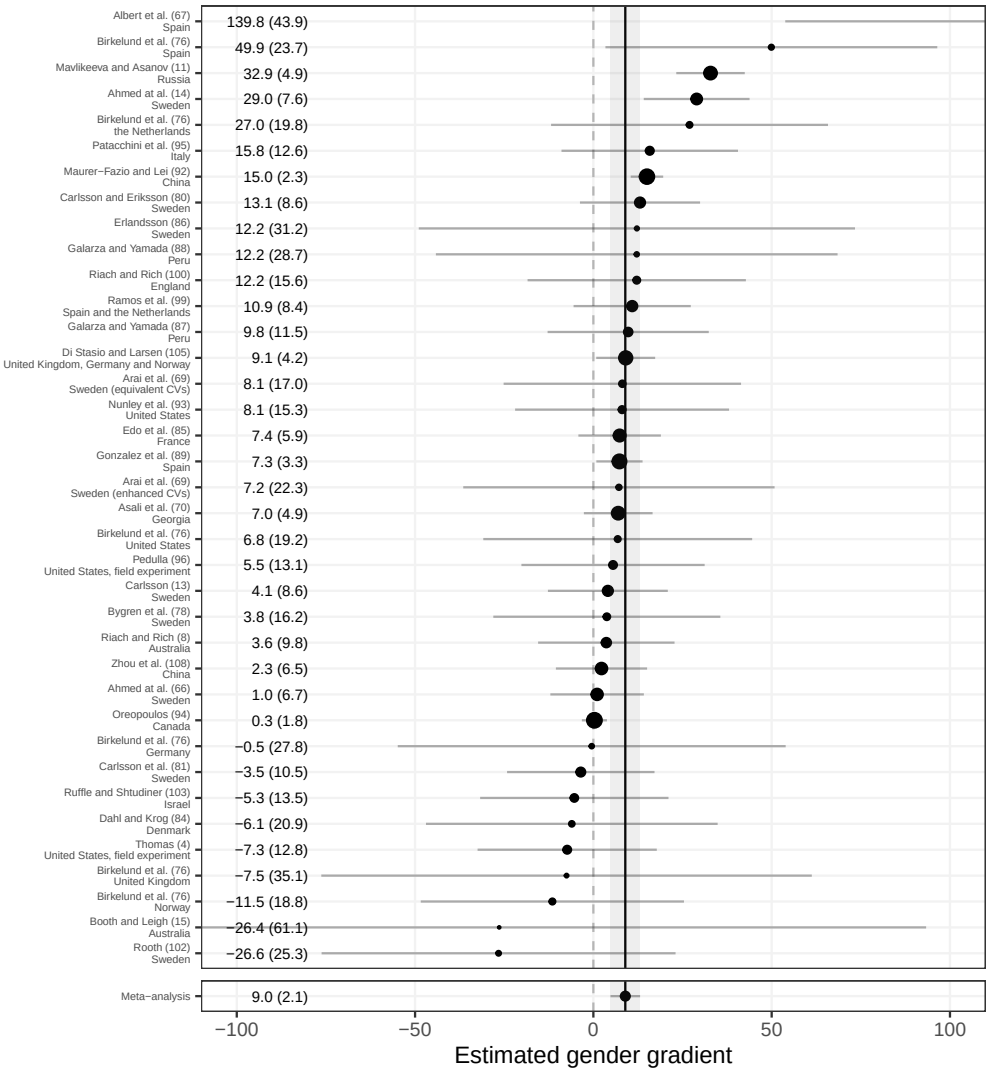


Figure 3: Figure S5.62 as reproduced by the maintained rewrite.

9 Maintained rewrite

The rewrite lives in `maintained/`: `helpers.R` plus fifteen scripts covering one cleaning step, three published tables, eight figures, two sets of in-text quantities, and the claims file that recomputes every number the article states. It is a translation, not a reanalysis. Every estimator, specification and sample restriction is the one the paper used, and the published tables come out of `metafor::rma.uni()` with the same formulas the archive writes.

9.1 Architecture

Table 9: The maintained rewrite.

Script	Produces
helpers.R	Packages, paths, formatting helpers
clean_gender_gradients.R	gender_gradients.csv
table_1_meta_regression.R	table_1_meta_regression.tex and .csv
table_2_ethnicity_meta_regression.R	table_2_ethnicity_meta_regression.tex and .csv
table_s2_1_study_manifest.R	table_s2_1_study_manifest.tex and .csv
figure_4_cates_gender_gradient.R	Figure 4
figure_5_cates_ethnicity.R	Figure 5
figure_s1_1_field_ates_forest.R	Figure S1.1 and its 116 plotted values
figure_s2_2_evidence_map.R	Figure S2.2
figure_s3_3_survey_field_comparison.R	Figure S3.3 and the two fitted gradients
figure_s3_4_survey_ates_forest.R	Figure S3.4 and its 26 plotted values
figure_s4_study_panels.R	Figures S4.5 to S4.61 and their 335 plotted values
figure_s5_62_gender_gradient_forest.R	Figure S5.62 and its 76 plotted values
text_summary_stats.R	The counts quoted in the text and in section S5
text_difference_in_magnitude.R	The predicted CATEs and the bootstrapped SE
in_text_claims.R	One printed line per claim the article makes

Four decisions are worth stating.

The study-level gender gradients are recomputed, not read. `clean_gender_gradients.R` refits the 37 within-study meta-regressions from the occupation CATEs rather than loading the deposited `gender_gradients_df.rds`. Depending on that file would mean depending on an intermediate the deposited code cannot regenerate. The two agree to 4e-13.

The uncovered floats are covered. `figure_s4_study_panels.R` and `table_s2_1_study_manifest.R` rebuild the 57 study panels and the study manifest, neither of which the deposit produces. Rebuilding them is what makes the gaps in the deposit measurable: without a reconstruction there is nothing for the 335 panel labels and the 171 manifest cells to be compared against.

Every value that appears in `output/` is computed from the deposited data. No published number is typed into a script as an input to a calculation, and the figures write their plotted labels to CSV so that a value shown in a figure can be traced to the fit that produced it rather than to a transcription.

The bootstrap runs at 500 resamples. That is the cost of the published standard error and it is paid rather than avoided. `rsample::bootstraps()` at `set.seed(343)` gives the same resamples as the archive's `sample_n()` loop, so the two agree exactly at 0.5781.

9.2 Patterns replaced

Table 10: Patterns replaced in the maintained rewrite.

Original pattern	Replacement	Reason
magrittr pipe	native pipe	House style
‘rm(list = ls())‘	(omitted)	House style
relative paths from the archive root	‘here::here()‘	Scripts must run from the project root
‘texreg()‘ printed to the console	‘modelsummary(output = ...)‘ to ‘.tex‘, plus ‘write_csv()‘	Console-only output cannot be diffed between runs
‘size =‘ in ‘geom_hline()‘	‘linewidth =‘	Deprecated in ggplot2 3.4.0; warns on every run
‘fct_reorder()‘ on an NA sort key	replace the key with ‘-Inf‘ before reordering	Errors in forcats 1.0.0
‘legend.position = c(x, y)‘	‘legend.position = ”inside”‘ with ‘legend.position.inside‘	House style; the numeric form still works
‘sims <- 5‘ with a comment to raise it	‘sims <- 500‘	Recovers the published standard error
manual ‘sample_n()‘ bootstrap loop	‘rsample::bootstraps()‘	Same resamples at the same seed, in a maintained package
‘predict.rma()‘ called directly	‘predict()‘ with S3 dispatch	Method called by name rather than through the generic

`xtable` is loaded by the archive but never called, so it is simply dropped. `texreg` still installs and still works; it is replaced because it writes only to the console, which means the published tables exist nowhere as files and cannot be diffed between runs.

10 R environment

Item	Value
R version	4.6.2
Platform	aarch64-apple-darwin23

Table 12: Package versions used for the run behind this report.

Package	Version
tidyverse	2.0.0
metafor	5.2.1
broom	1.0.13
scales	1.4.0
rsample	1.3.2
modelsummary	2.6.0
knitr	1.52

kableExtra	1.4.1
here	1.0.2